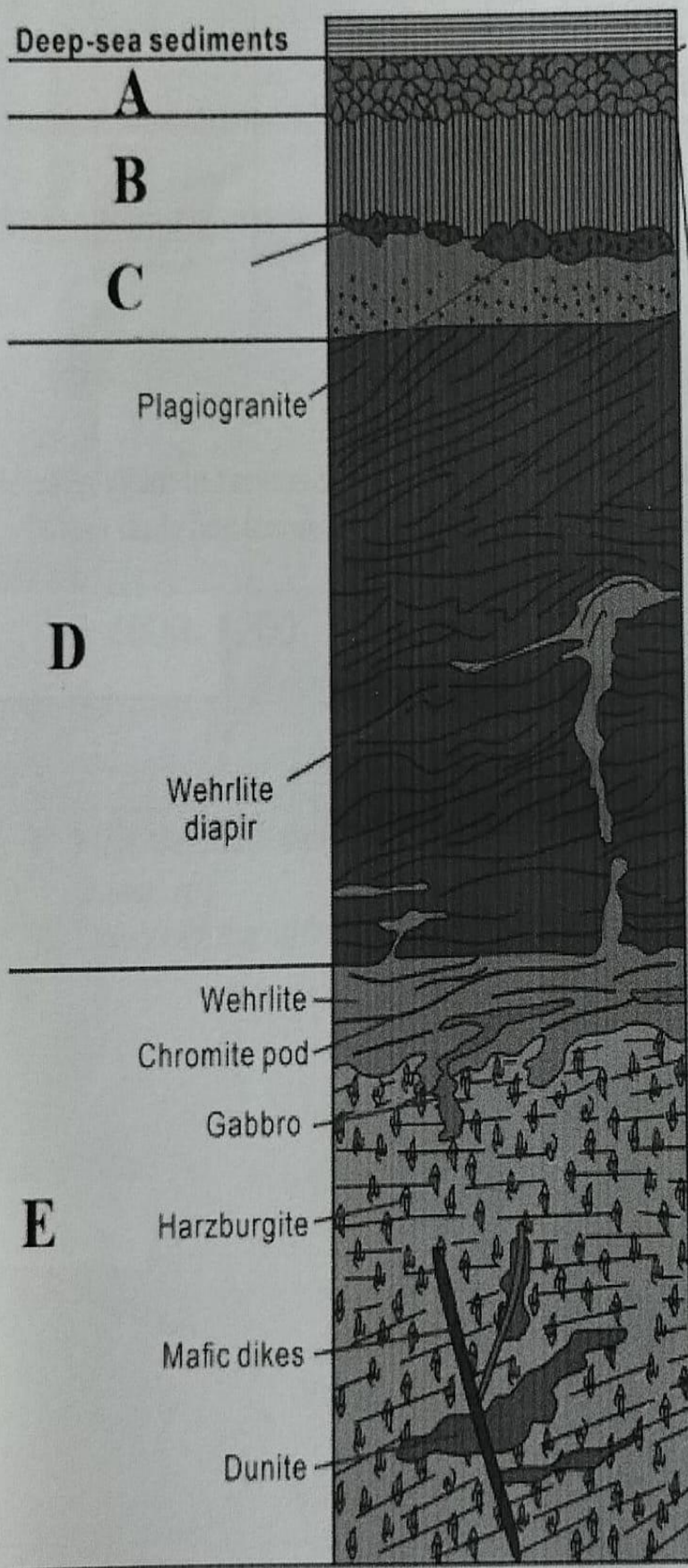
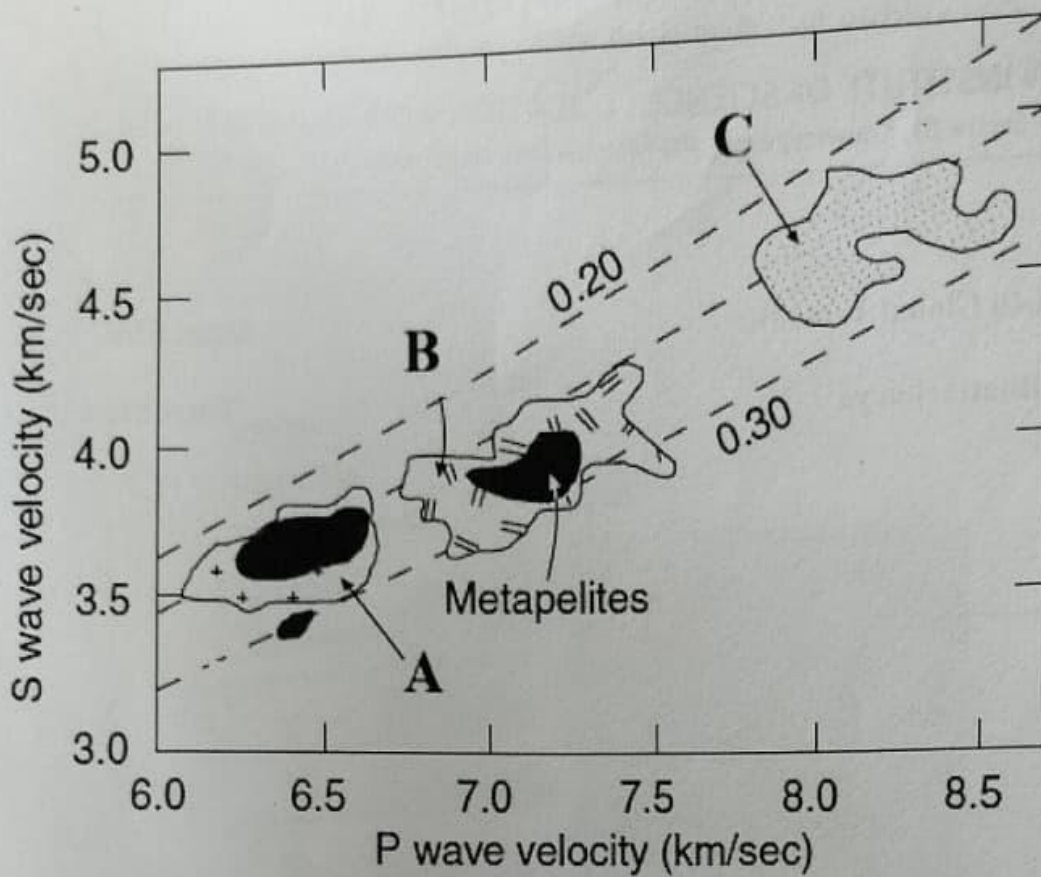
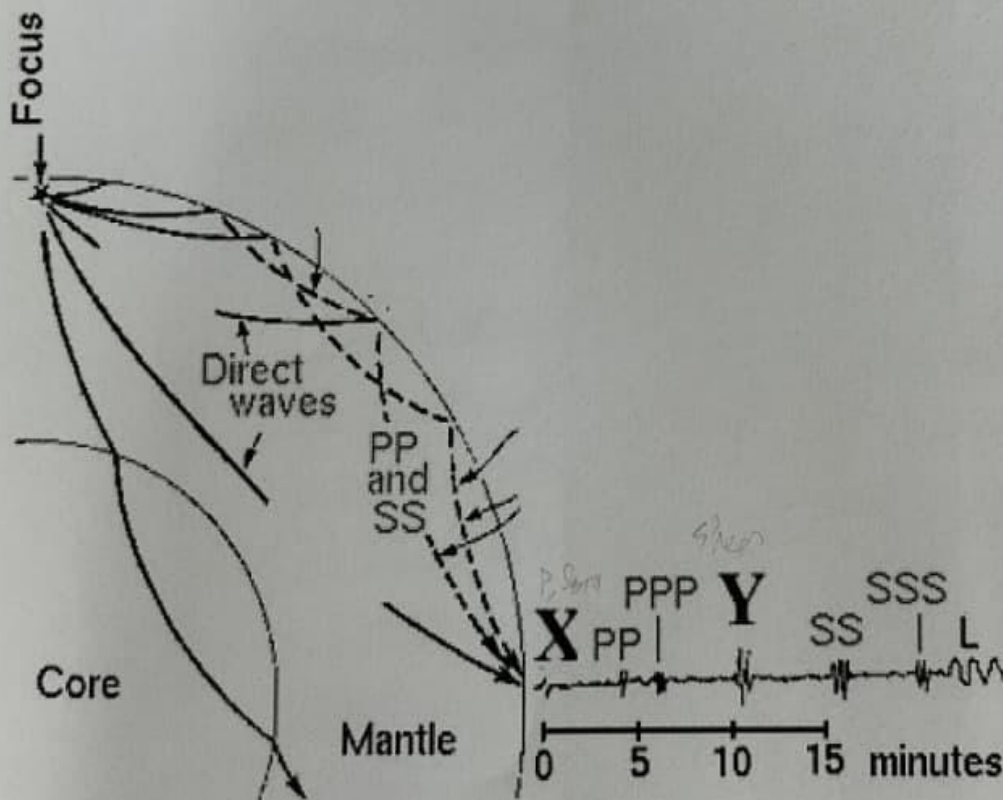




- 1) Identify the lithological units A to D in the given image of a suprasubduction zone type ophiolite (ophiolites are obducted oceanic crust). What kind of rock types does E indicate? Explain the formation of A and D. (8)

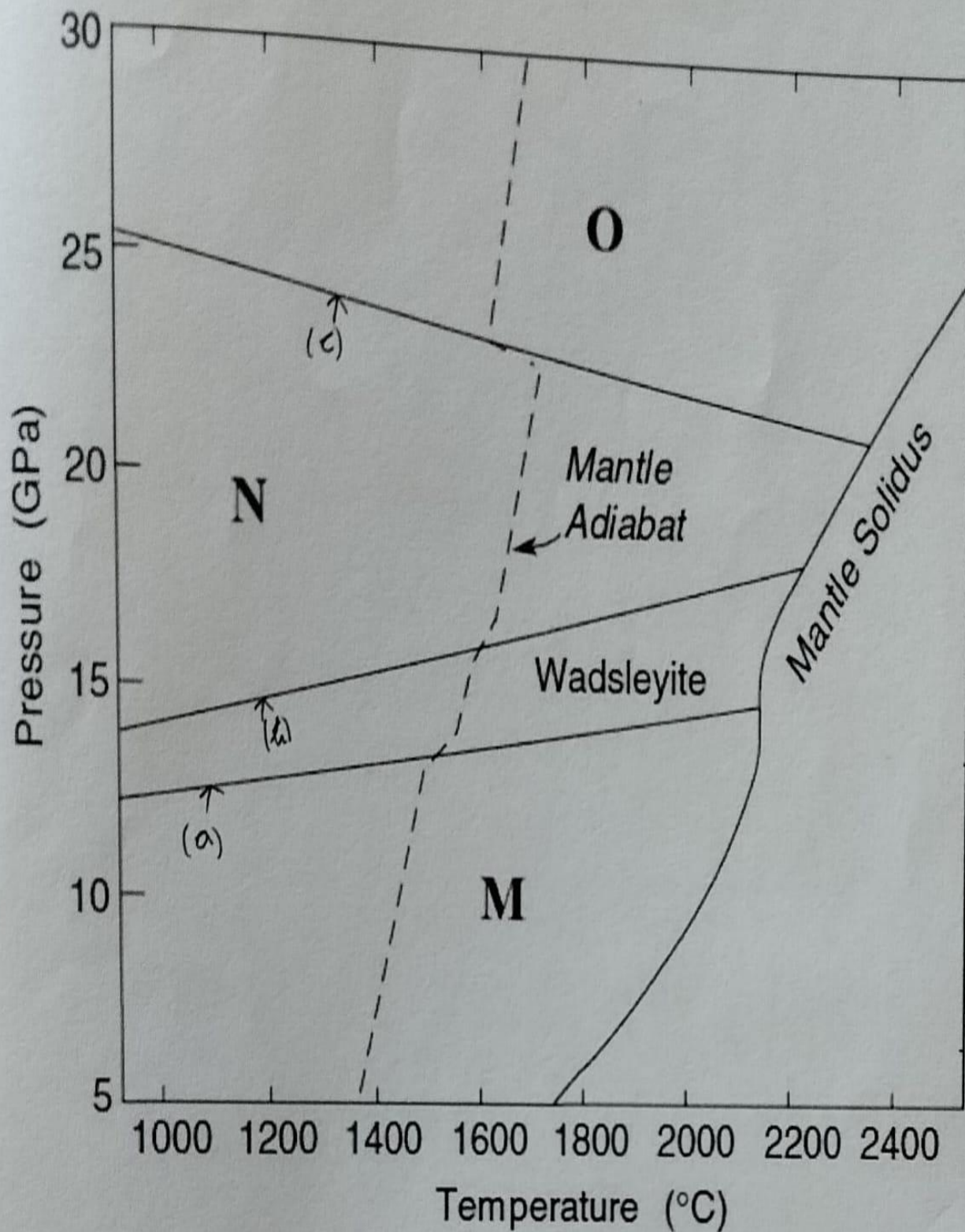


- 2) Refer to the image above and identify the basic rock types in the context of mafic minerals content marked as A, B, C using your understanding of compressional and shear wave velocities. (4.5)



- 3) Arrival of direct and reflected body waves on seismograms are seen in the given diagram. Identify the direct waves X and Y with proper explanations. (4)





- 4) Mark the phase transition discontinuities in the given image above. Identify M, N and O phases. (6)
- 5) Enumerate the differences between continental and oceanic crusts. (2.5)